

Regression, Bias–Variance Dilemma & Model Selection

Continuous Output Prediction and Optimal Model Complexity

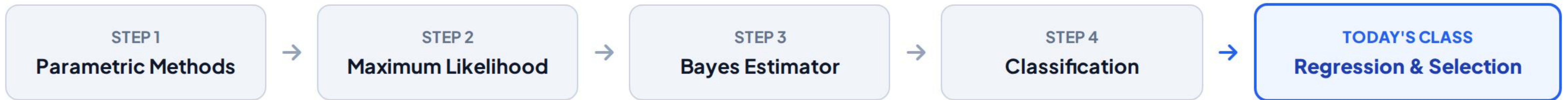
Reference: Ethem Alpaydin, *Introduction to Machine Learning* (MIT Press/PHI)

Coverage: Chapter 4 (Sections 4.6, 4.7, 4.8)



Teacher Introduction: "Welcome students. Today is our concluding lecture for Unit III. We are moving from classifying discrete categories to predicting real numbers and learning how to choose the right model."

Connecting With Previous Classes



"In earlier lectures, we estimated density parameters to decide **which class** an object belongs to. Today, we use parametric models to **predict continuous numerical values** and discover how to pick the model with just the right complexity."



Teacher Explanation: Remind students that MLE gave us means and variances for classification. Now we apply those same parameter-fitting principles to predict continuous quantities.

What is Regression?

Classification

Output is a **category or discrete class label**.

- ⦿ Email → Spam or Not Spam
- ⦿ Tumor → Benign or Malignant
- ⦿ Image → Cat, Dog, or Bird

Regression

Output is a **continuous numerical value**.

- ✓ House Size (sq.ft) → Price in ₹
- ✓ Hours Studied → Exam Marks (0–100)
- ✓ Years of Experience → Annual Salary

Formal Definition: Regression models the conditional expectation of a real-valued output variable y given input x .

 **Teacher Explanation:** "Ask students: 'If the answer is a category like Pass/Fail, what is it?' Classification. 'If the answer is 78.5 marks, what is it?' Regression."

Basic Regression Example: Study vs. Marks

Hours Studied (x)	Actual Marks (y)
1	35
2	45
3	55
4	65
5	75

Notation Used in Textbook:

x = **Input variable** (feature/attribute, e.g., study hours)

y = **Actual true output** (target value, e.g., real marks)

$\hat{y}$ = **Predicted output** (computed by our machine learning model)

Goal: Learn a mathematical mapping function $g(x|\theta)$ so that when a student studies for 6 hours, we can accurately predict their marks $\hat{y}$.

Linear Regression: The Straight Line Model

$$\hat{y} = b_0 + b_1x$$

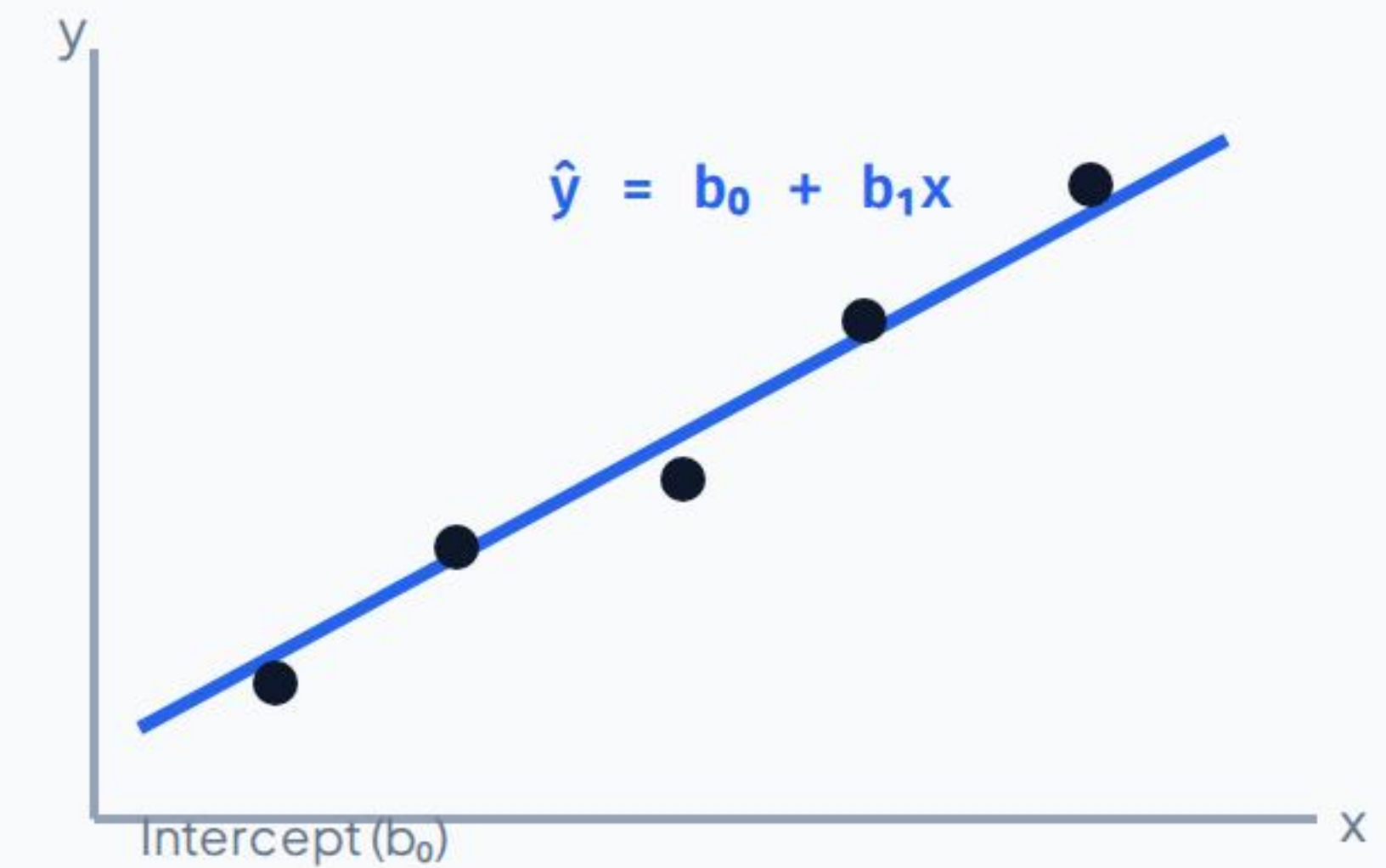
Meaning of Each Symbol:

$\hat{y}$: Predicted continuous output value

x : Given input variable (predictor)

b_0 : **Intercept** (predicted value of y when $x = 0$)

b_1 : **Slope / Weight** (how much $\hat{y}$ changes when x increases by 1)



Linear regression finds the straight line that best fits the data points.

How Linear Regression Makes Predictions

Given Learned Parameters:

Intercept: $b_0 = 25$ (Base marks)

Slope: $b_1 = 10$ (Marks per study hour)

Fitted Model: $\hat{y} = 25 + 10x$

Predicting for a New Student ($x = 6$ hours):

Step 1: $\hat{y} = 25 + 10(6)$

Step 2: $\hat{y} = 25 + 60$

Result: $\hat{y} = 85$ marks

Takeaway: The model replaces human guesswork with mathematical prediction based on learned training parameters.



Teacher Explanation: "Demonstrate how simple this is. Once the parameters b_0 and b_1 are learned, predicting output for any new x is just plugging in the number."

Residual / Prediction Error in Regression

Residual Equation:

$$e_i = y_i - \hat{y}_i$$

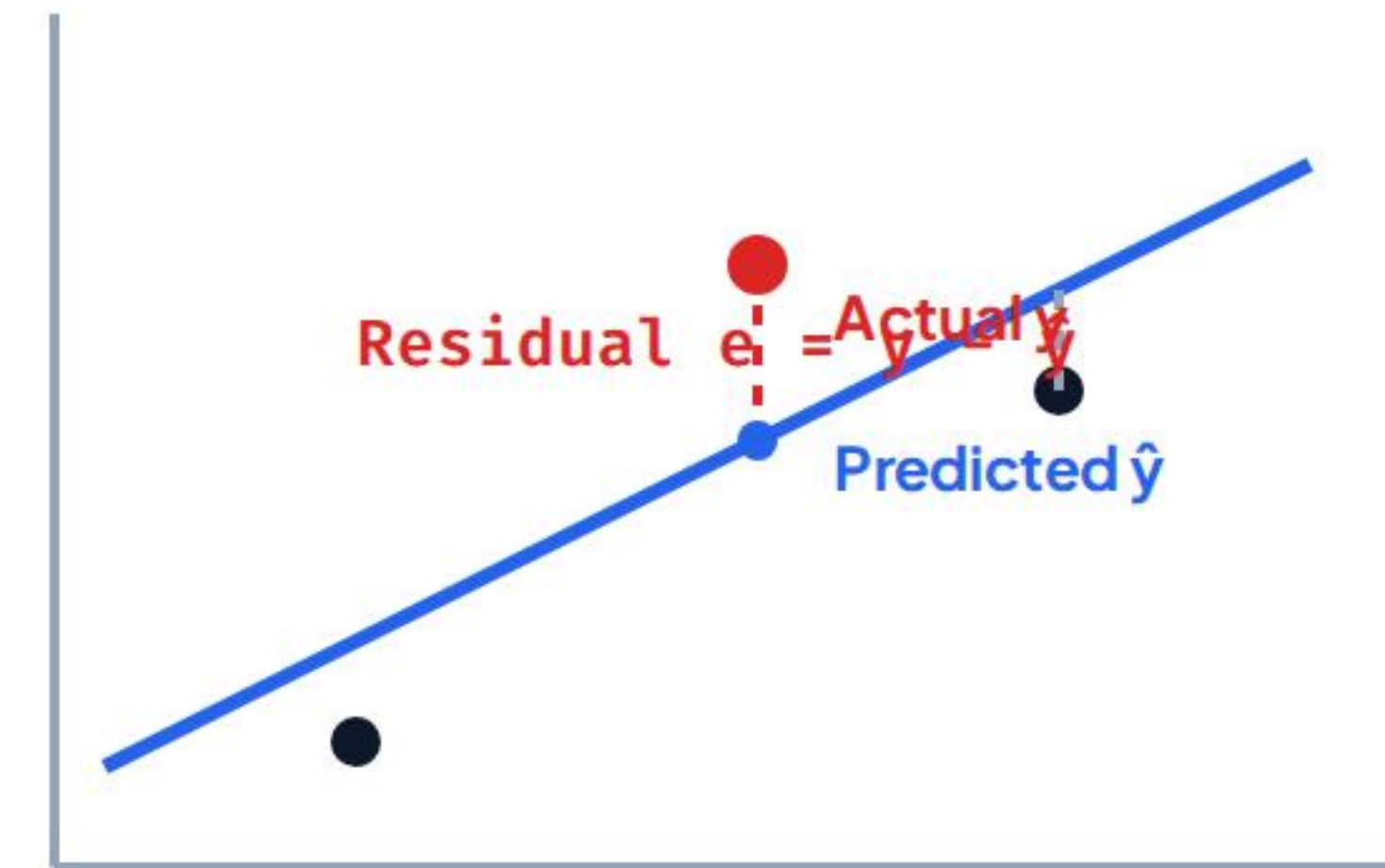
Error = Actual Value (y) – Predicted Value ($\hat{y}$)

Concrete Example:

If a student actually scored $y = 80$, but our regression model predicted $\hat{y} = 75$:

$$e = 80 - 75 = +5 \text{ marks error}$$

Objective of Regression: Find parameters b_0, b_1 that minimize the total sum of squared residuals across all training examples.



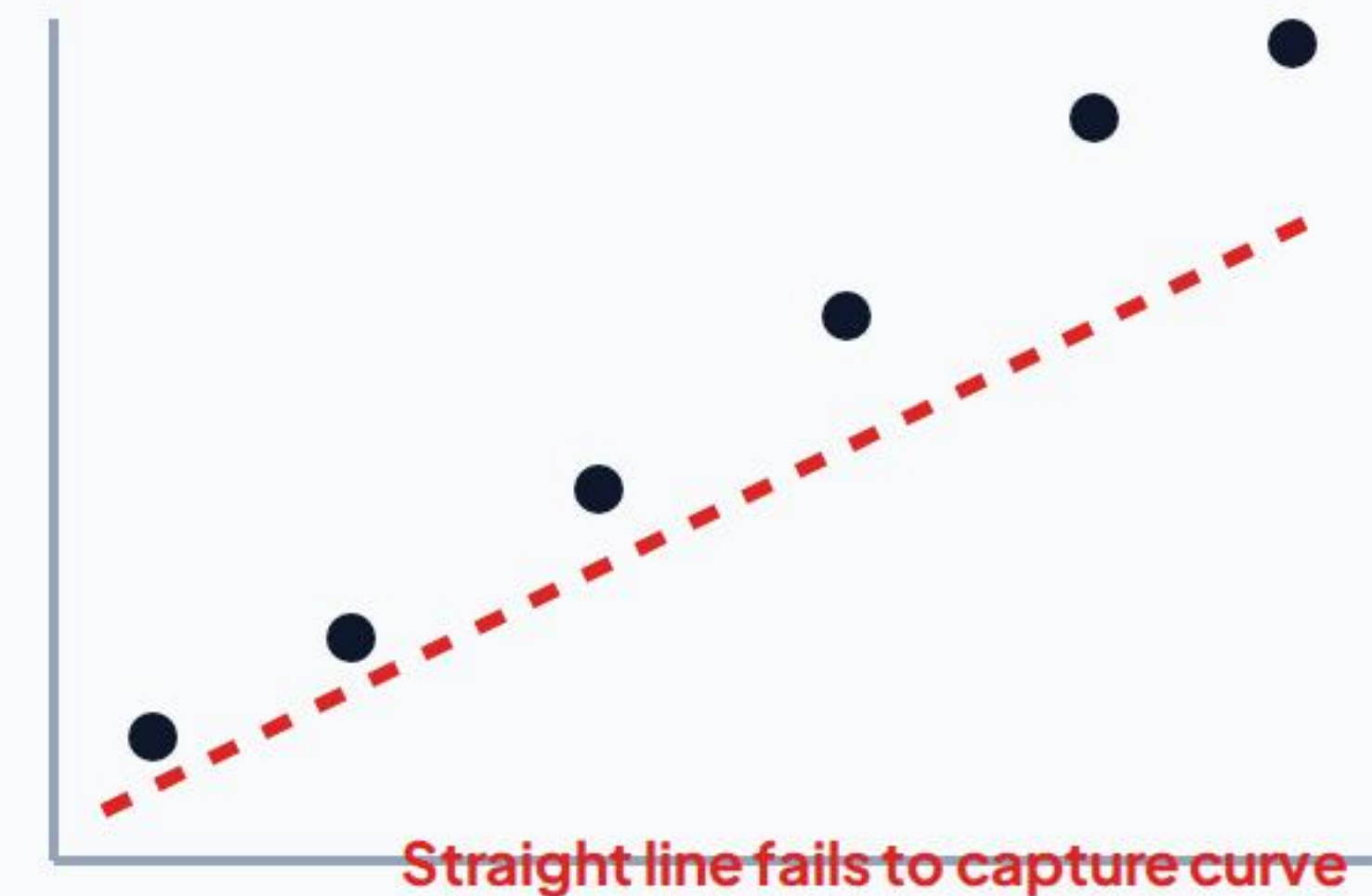
The vertical dashed distance is the residual (prediction error).

Why a Straight Line is Not Always Enough

Limitation of Linear Models:

Real-world relationships in nature and business are rarely perfectly straight.

- ⚠ If data curves upward or saturates, a straight line will systematically miscalculate.
- ⚠ Forcing a straight line onto curved data causes severe structural error.
- **Solution:** We need a more flexible model that can bend with the data.



The data is quadratic/curved, but the straight line is rigid and inaccurate.



Teacher Explanation: "Tell students: 'Think of human growth or car braking distance. Braking distance does not double when speed doubles—it quadruples! A straight line cannot capture that.'"

Polynomial Regression: Modeling Curves

Polynomial regression adds powers of the input x (x^2 , x^3 , ...) to allow the fitted function to bend:

DEGREE 1 (Linear)

$$\hat{y} = b_0 + b_1x$$

A straight line. Rigid with no curves.

DEGREE 2 (Quadratic)

$$\hat{y} = b_0 + b_1x + b_2x^2$$

A parabola. Allows one single bend/curve.

DEGREE 3 (Cubic)

$$\hat{y} = b_0 + b_1x + b_2x^2 + b_3x^3$$

Allows two bends for complex variations.

Important Insight from Alpaydin: Even though it produces a curved line, it is still called a *linear model* because it is linear in the parameters $b_0, b_1, b_2 \dots$!



Teacher Explanation: "Highlight that higher degree polynomials add more flexibility. But ask the students: 'Can we just use Degree 20 to make it super flexible?' That leads to model complexity!"

What is Model Complexity?

Model Complexity describes how *flexible* a model is and how many parameters it has to fit data.

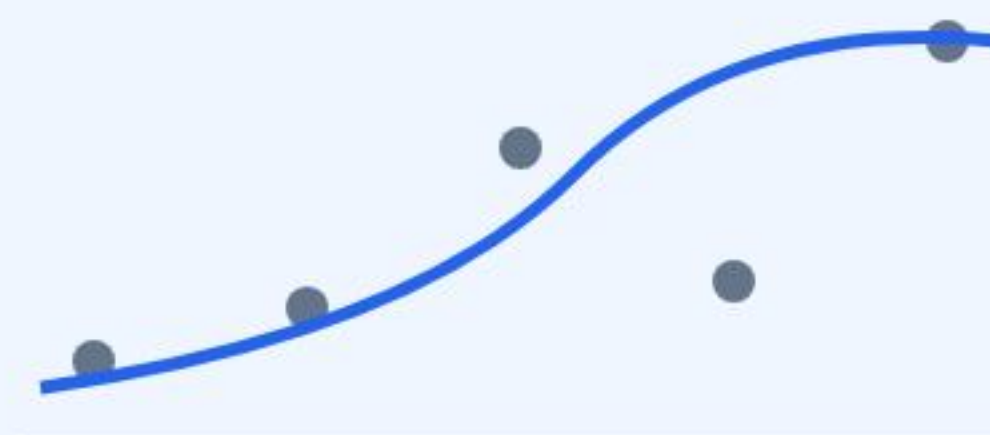
LOW COMPLEXITY



Degree 1 (Linear)

Rigid, few parameters, low flexibility.

BALANCED COMPLEXITY



Degree 2 or 3

Smooth curve, captures true shape.

VERY HIGH COMPLEXITY



Degree 8+ (High-Order)

Hyper-flexible, erratic oscillations.



Teacher Explanation: "Use the rubber band analogy: Low complexity is a rigid steel rod; moderate complexity is a supportive flexible ruler; high complexity is a loose cooked noodle."

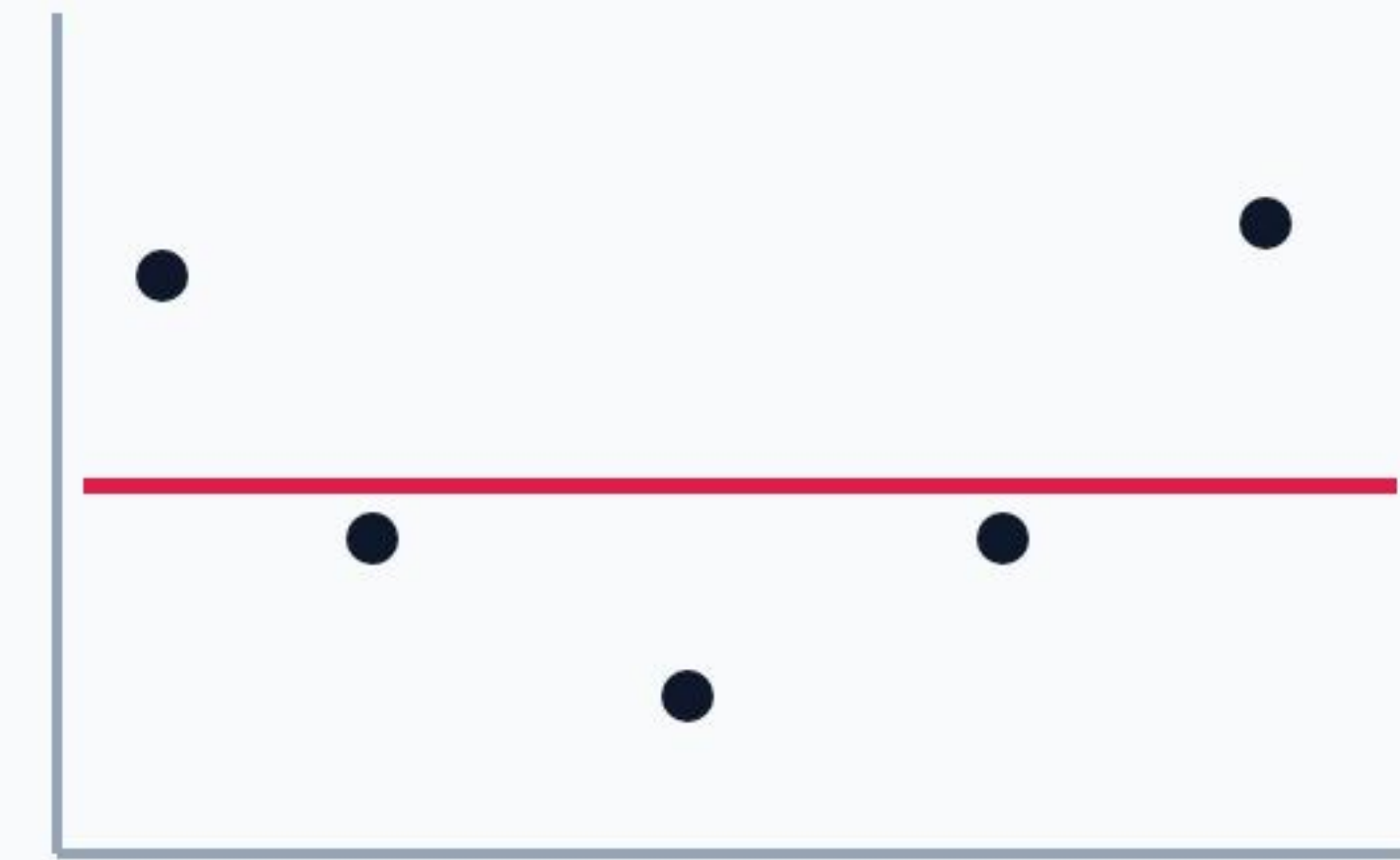
Underfitting & High Bias

Underfitting: Model Is Too Simple

Occurs when a model lacks the capacity to represent the underlying true pattern of the data.

- ✗ **High Bias:** Strong rigid assumptions that ignore data reality.
- ✗ **High Training Error:** Performs poorly even on the data used to train it.
- ✗ **High Test Error:** Also fails on unseen new data.

Key Chain: Too Simple Model → High Bias → Underfitting



Straight line cannot fit a curved bowl pattern!



Teacher Explanation: "Underfitting is like a student who only memorized one single formula for the entire syllabus. No matter what question comes, they write that one formula."

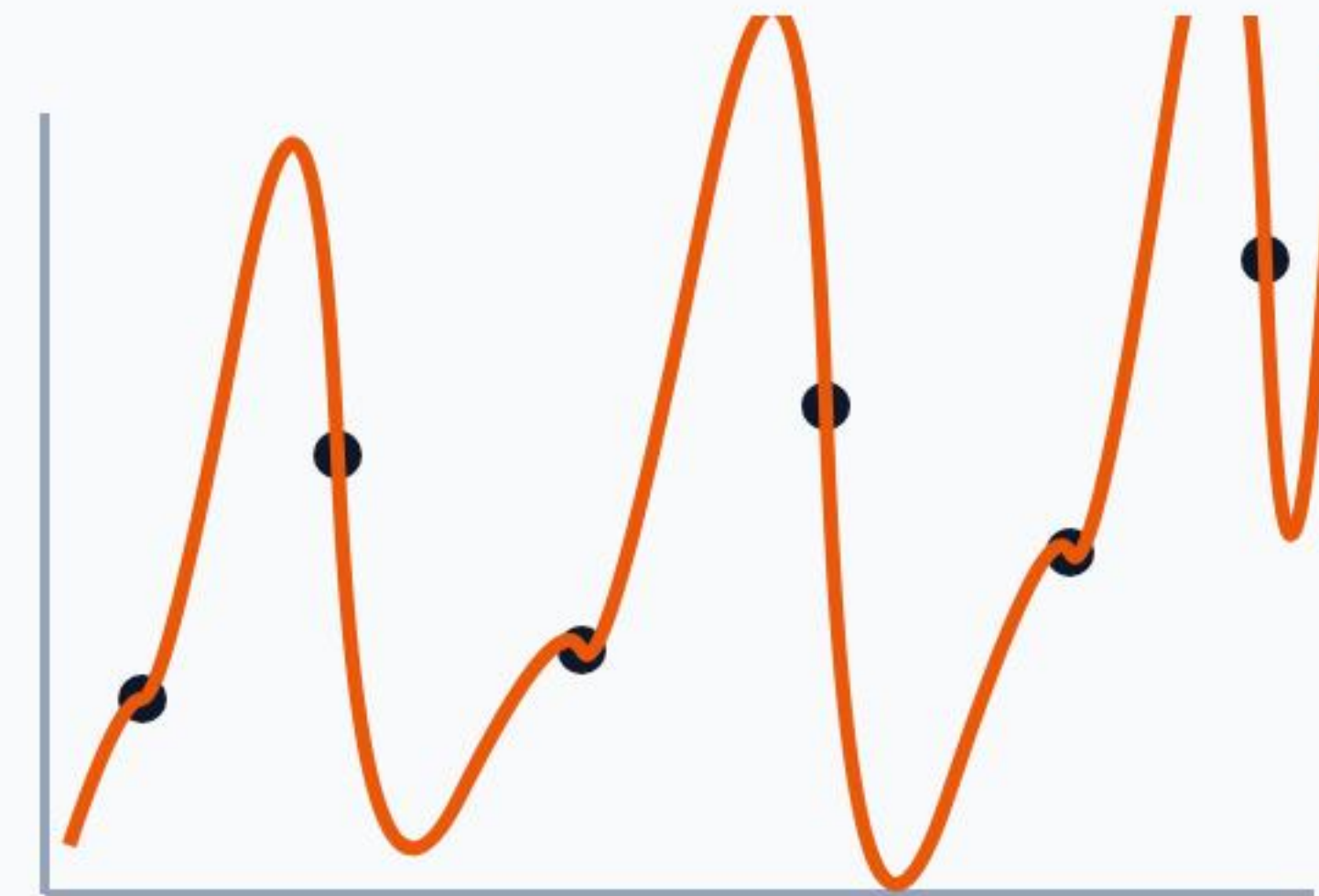
Overfitting & High Variance

Overfitting: Model Memorizes Noise

Occurs when the model is so flexible that it memorizes the training data points instead of learning general rules.

- ⚠️ **Zero / Low Training Error:** Passes perfectly through every training dot.
- ⚠️ **High Variance:** Highly sensitive to minor changes in training samples.
- ⚠️ **Catastrophic Generalization:** Fails miserably on new, unseen test data.

Key Chain: Too Complex Model → High Variance → Overfitting



Crazy wavy curve passes through noise but has zero predictive value!

Deconstructing Bias and Variance

Bias

"How far the model's average prediction across different training sets is from the true underlying relationship."

High Bias = Inflexible model, oversimplified assumptions.

Variance

"How much the learned model changes and swings when trained on a different training dataset."

High Variance = Unstable model, overly sensitive to data sample.

Crucial Rule from Alpaydin: We CANNOT simply set both Bias = 0 and Variance = 0. They pull in opposite directions!



Teacher Explanation: "Remind students of Section 4.3 from last week where we computed variance of estimators. Now we apply it to entire prediction functions."

The Bias–Variance Trade–Off

The **Bias–Variance Dilemma** is the inevitable trade-off when selecting model complexity:

As Complexity Increases:

- Bias decreases (model fits data better)
- Variance increases (model becomes erratic)

$$\text{Total Expected Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}$$

Optimal Goal: Find the "sweet spot" at the bottom of the U-curve where Total Error is minimized!



Training Error vs. Generalization

Comparing Three Candidate Models:

Model	Training Error	Unseen Test Error
Model A (Deg 1)	10.2	11.0
Model B (Deg 3)	4.8	5.1 (BEST)
Model C (Deg 10)	0.0 (Zero!)	34.5 (FAIL)

Model C has 0 training error because it memorized the training set!

FUNDAMENTAL DEFINITION

Generalization

"Generalization is the ability of a trained machine learning model to perform accurately on **unseen, novel data**."

Key Exam Principle: A lower training error does *NOT* mean a better model!
Never pick a model based purely on training error.



Teacher Explanation: "Tell the class: 'If a student scores 100/100 on textbook exercises because they had the answer key, will they definitely pass the final university exam?' No!"

Why Do We Need Model Selection?

The Practical Question:

"We have several candidate polynomial models: Degree 1, Degree 2, Degree 3, Degree 4, Degree 5... **Which one should we deploy in the real world?**"

❌ WRONG APPROACH

Picking the model with lowest training error always selects the highest polynomial degree, which causes severe overfitting!

✓ SCIENTIFIC APPROACH

Use formal **Model Selection Procedures** to find the model that optimizes generalization performance.

Definition (Alpaydin): Model selection is the process of choosing an appropriate model of right complexity from a hypothesis class.

 **Teacher Explanation:** "Model selection is like goldilocks: Not too simple (underfit), not too complex (overfit), but just right."

Training Set vs. Validation Set

Training Set (e.g. 80 Observations)

Validation (20)

✂ Training Data (80%)

Used directly by the learning algorithm to calculate the numerical parameters (b_0 , b_1 , $b_2 \dots$).

⚖ Validation Data (20%)

Kept strictly hidden during parameter training. Used only to test candidate models and select the best complexity!

Never evaluate a model on the same data it learned from. That would be like grading an exam using the practice worksheet!

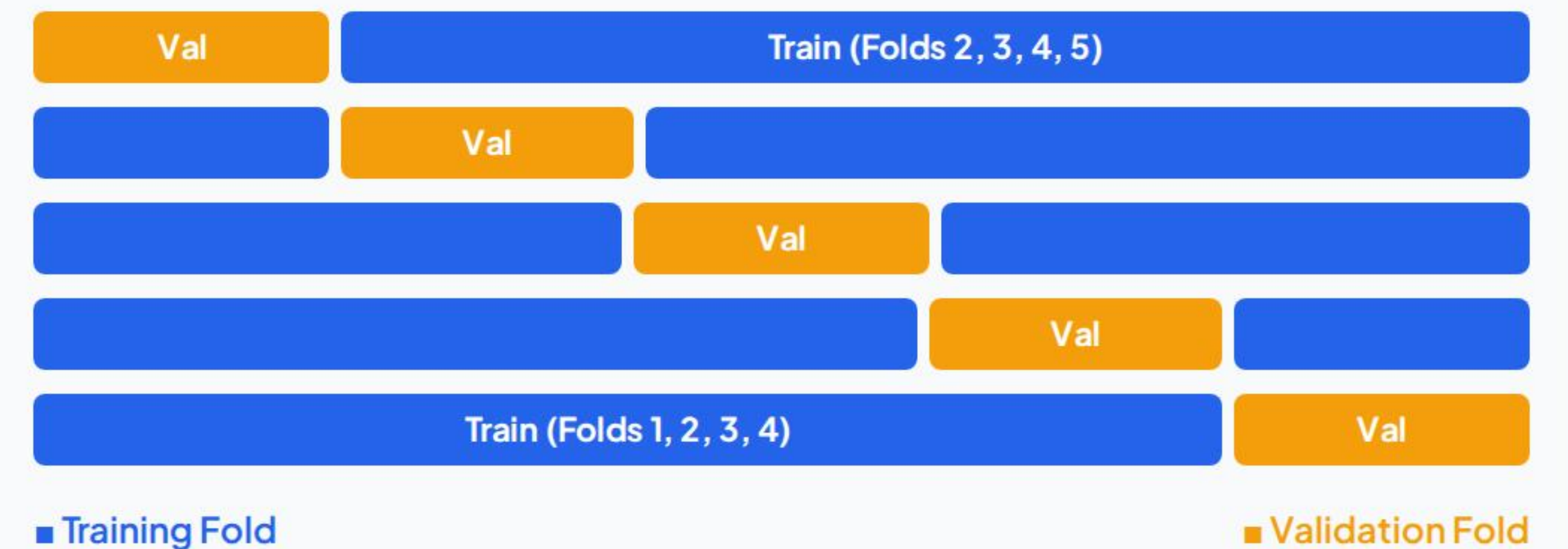
 **Teacher Explanation:** "Explain that validation data acts as a mock exam before deploying to real unseen test scenarios."

K-Fold Cross-Validation

When dataset size is small, a single train/validation split is wasteful. **K-fold Cross-Validation** rotates validation:

1. Divide entire dataset into **K** equal parts (folds).
2. Hold out 1 fold for validation; train on remaining **K - 1** folds.
3. Repeat **K** times so every fold acts as validation exactly once.
4. Compute the **average validation error** across all **K** iterations.
5. Select the model degree with lowest average validation error.

5-Fold Cross Validation Layout:



Selecting the Best Polynomial Degree

Candidate Model	Train Error	Validation Error	Diagnosis
Degree 1 (Linear)	14.2	12.0	Underfitting
Degree 2 (Quadratic)	6.5	7.0	Good
Degree 3 (Cubic)	4.2	5.0 (LOWEST!)	SELECTED
Degree 4 (Quartic)	1.8	11.0	Overfitting Starts

Winning Model: Degree 3

Validation error reaches its minimum (5.0) at Degree 3. At Degree 4, validation error starts shooting back up, signalling overfitting.

Note: This is an illustrative academic example demonstrating the textbook procedure for finding the optimal polynomial order.

Other Model Selection Procedures

Besides cross-validation, Alpaydin discusses theoretical penalty and information criteria:

Regularization

Adds a penalty to the loss function to keep parameter values small and avoid wild curves.

AIC & BIC Criteria

Statistical formulas that balance goodness-of-fit with a mathematical penalty for number of parameters.

Minimum Description Length (MDL)

Prefers the model that requires the shortest bit-length to encode both the model and its residuals.

Structural Risk Minimization (SRM)

Selects a model that minimizes an upper bound on generalization error using VC-dimension.

Note for BCA Exam: Cross-Validation is the primary practical method tested in university exams.



Teacher Explanation: "Reassure students: They do not need deep math derivations for AIC/BIC/MDL. Remembering their one-line definitions is sufficient for the university exam."

The Complete Parametric Machine Learning Loop

ESTIMATION
MLE & Bayes



TASKS
Classify or Regress



DILEMMA
Bias vs. Variance



SELECTION
Cross-Validation



GOAL
Generalization

The Unified Lesson of Chapter 4:

"We choose a parametric model, estimate its unknown parameters using maximum likelihood or Bayesian methods, evaluate its bias and variance, and use validation to pick the exact complexity that guarantees high performance on unseen data."

Teacher Explanation: "Wrap up Unit III by tying every chapter section together into this single unified 5-step engineering framework."

Expected University Exam Questions

Short Answer (2–3 Marks)

1. Define regression with an example.
2. Differentiate classification and regression.
3. Write the linear regression formula and define symbols.
4. What is model complexity?
5. Define underfitting and overfitting.
6. What is the bias-variance dilemma?
7. Define generalization.
8. What is K-fold cross-validation?

Descriptive (5–10 Marks)

1. Explain linear regression and show how predictions are made using an example.
2. Explain polynomial regression and how model complexity relates to polynomial degree.
3. Discuss the Bias-Variance dilemma with a suitable error vs. complexity diagram.
4. Explain model selection using K-fold cross validation with a neat diagram.
5. Write short notes on: Regularization, AIC/BIC, MDL, and SRM (Section 4.8).



Teacher Explanation: "Tell students: 'Questions 3 and 4 under descriptive are repeated almost every single semester in university papers! Prepare those diagrams carefully.'"

Unit III Final Cheat Sheet

Regression

Predict continuous numerical values.

Linear Model

Straight line: $\hat{y} = b_0 + b_1x$

Polynomial

Adds powers (x^2 , x^3) for curves.

Complexity

Flexibility to fit diverse shapes.

Underfitting

Too simple → High Bias.

Overfitting

Too complex → High Variance.

Dilemma

Balance bias and variance on U-curve.

Cross-Validation

Rotate K-folds to find best model.